

Requirements List

Course: Product Development – From Need to Market

Institution: University of applied sciences Kaiserslautern

Semester: SS 2025

Intorduction

The following specification outlines the functional and non-functional requirements for the design and development of the robotic equipment described herein. It forms the basis for design decisions, validation, and verification activities.

This document is subject to controlled updates throughout the lifecycle of the project. Any modification must not compromise equipment safety, operational effectiveness, or engineering soundness.

If a requirement does not explicitly list a new date of change, it is to be understood that the date of change is the same as the requirement listed immediately above it.

Abbreviations

HMI	Human-Machine Interface
ROI	Return on Investment
GS	Geprüfte Sicherheit
CE	Conformité Européenne

Legend

F	Fixed requirement
G	Goal requirement
D	Desired requirement

Position	Date of Change	Type of requirement	Requirements List “Autonomous Robot for Invasive Plant Removal”	Created 08.04.2025
			Goal / main function: To autonomously identify, locate, and remove invasive plants in natural environments without harming native vegetation.	Responsible
1	12.04.25		Area of Application	
1.1		F	Suitable for meadows, pastures, and light forest terrain.	
1.2		G	Capable of operating in areas with vegetation cover including bushes and sparse trees.	
1.3		G	Able to overcome common terrain obstacles (stones, bumps, ditches up to 25 cm depth/height).	
		F	Capable of driving through grass and light bushes without performance loss.	
2	12.04.25		Target Plant Types	
2.1		F	Recognizes invasive species such as: perennial lupine, giant hogweed, common ragwort.	
2.2		F	Operates with an AI-based plant recognition system (visual detection with plant database).	
2.3		G	Identification accuracy >= 95%	
2.4	14.04.25	G	Target plants: Height 30–150 cm, stem diameter up to 5 cm.	
2.5		G	Roots to be targeted down to approx. 20–30 cm depth.	
3	12.04.25		Removal Method / Effectiveness	
3.1		F	Effectively removes target plant to prevent regrowth (e.g., root removal or neutralization).	
3.2		G	Regrowth prevention effectiveness >= 90% per treated plant.	
3.3		F	Does not damage nearby plants or surrounding soil (non-target impact < 5%).	
4	12.04.25		Mobility and Terrain Compatibility	
4.1		F	Low soil compaction: max ground pressure <= 30 kPa.	
4.2		G	Usable on slightly damp soils without traction loss.	
4.3		G	Operational slope tolerance: up to 20° incline.	

5	12.04.25		Power Supply / Operation Time	
5.1		F	Autonomous operation without grid connection.	
5.2		G	Continuous working time ≥ 6 hours per charge/fuel cycle.	
5.3		G	Recharge or refuel time ≤ 2 hours.	
5.4	14.04.25	F	Electrical operation: 230V, 50Hz supply for charging.	
5.5		G	Connected electrical load ≤ 1.5 kW.	
6	12.04.25		Safety	
6.1		F	Emergency stop feature accessible and clearly marked.	
6.2		F	Sensors to detect human presence and stop operation automatically.	
6.3		F	Safety perimeter monitoring for unauthorized access.	
6.4	14.04.25	F	Visual alerts when in operation.	
6.5		F	Must detect persons within a radius of at least 5 meters.	
6.6		F	Audible alerts must activate when the working zone is penetrated.	
6.7		F	GS-Certification.	
6.8		F	CE labeling compliant.	
7			Environmental Impact	
7.1	12.04.25	F	Operational temperature range: -10°C to $+40^{\circ}\text{C}$.	
7.2		G	Relative humidity: 10–90% non-condensing.	
7.3	14.04.25	G	Rain tolerant (IPX4 level minimum).	
7.4		D	Operation in light snowfall and moderate wind (<40 km/h).	
8			Usability and Handling	
8.1	12.04.25	F	Remote control or app interface for setup and monitoring.	
8.2		G	Interface language selection (See 15)	
8.3		D	Autonomy status feedback and progress visualization.	
9			Transport and Deployment	
9.1	14.04.25	F	Modular and compact form for transport.	
9.2		G	Deployment time < 15 minutes.	
9.3		F	Single person transport with assistance tool. Unit weight ≤ 80 kg.	

10			Cost Targets	
10.1	12.04.25	F	Maximum production cost: €15,000.	
10.2		G	Maintenance cost per year < €800.	
10.3		D	ROI time <= 5 years compared to conventional methods (cost of traditional/manual invasive plant removal methods).	
11			Service Life and Maintenance	
11.1	12.04.25	F	Minimum operational life: 5 years.	
11.2		G	Maintenance interval: every 200 hours of operation.	
11.3		D	Tool-less maintenance access.	
11.4		F	Cleaning duration: manually < 10 minutes.	
12			Recycling and Sustainability	
12.1	14.04.25	G	80% of materials recyclable.	
12.2	12.04.25	D	Eco-friendly packaging and logistics.	
13			Dimensions and Weight	
13.1	14.04.25	F	Max dimensions: 1.2 m (L) × 0.8 m (W) × 1.2 m (H).	
13.2		G	Max transport weight: 80 kg.	
14			Processing Speed	
14.1		F	Average removal time per plant <= 10 seconds.	
14.2		G	Minimum 6 plants removed per minute in dense vegetation.	
14.3		D	Feedback system to verify successful removal per plant.	
15			Languages	
15.1		F	Technical documentation in English.	
15.2		F	Source code documentation in English.	
15.3		F	Instruction manuals in English.	
15.4		G	Instruction manuals available in German, French, Spanish, Russian.	
15.5		F	HMI (screen/app) interface in English.	
15.6		G	HMI interface also available in German, French, Spanish, Russian.	
16			Project Timeline	

16.1	F	Project plan includes timeline, milestones, and documentation phases.	
16.2	F	Timeline to identify critical paths.	
16.3	F	Milestones must include commodity sourcing for testing with specified quantities.	
17		Documentation	
17.1	F	Functional specification including architecture, software, and hardware design with I/O list.	
17.2	F	Alarms test, critical and safety functions report.	
17.3	F	Safety component listing, hazard analysis, and design review.	
17.4	F	Equipment specification with configurable settings and parameters.	
17.5	F	Installation and re-assembly procedure.	
17.6	F	Controls documentation: switches, sensors, indicator functions.	
17.7	F	Mechanical adjustment and format changeover instructions.	
17.8	F	Cleaning and preventative maintenance instructions.	
17.9	F	Troubleshooting procedures.	
17.10	F	Bill of materials.	
17.11	F	Assembly drawings.	
17.12	F	Electrical schematics.	
17.13	F	Software copies of all programs	
18		Network	
18.1	F	All controllers with communication ports must include nameplate with IP, baud rate, serial config, etc.	
18.2	F	Wireless communication must comply with regulatory standards	
19		Warranty	
19.1	F	Minimum 1-year warranty covering all components	